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Form Approved
OMB No. 0704-0188

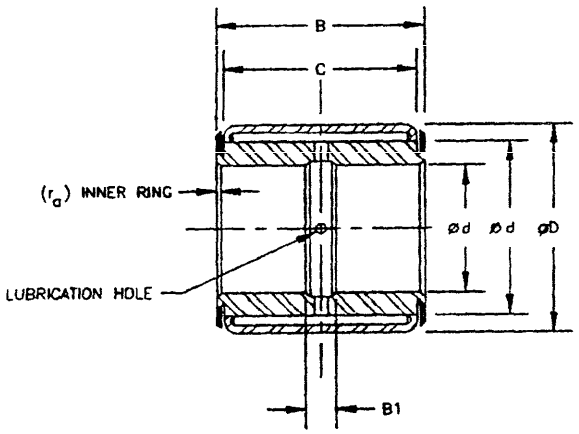



TABLE I — Dimensions in inches

| Dash | ød Bore | øD Outer Ring Outside Dia. 2/ | B Over-all Width | c Outer Ring Width | ød1 Washer Outside Dia. | B1 Lubri-cation Groove Width | $r_a$ 1/ Fillet Max. | Radial Internal Clearance Max. | Limit Load Rating Lbs 3/ | Mass (approx) lb |
|---|---|---|---|---|---|---|---|---|---|---|
| -3 | .1900 | .6250 | .625 | .500 | .562 | None | .022 | .0015 | 679 | .040 |
| -4 | .2500 | .6250 | .562 | .500 | .562 | None | .022 | .0015 | 878 | .025 |
| -5 | .3125 | .6875 | .625 | .500 | .625 | None | .022 | .0015 | 988 | .050 |
| -6 | .3750 | .7500 | .812 | .750 | .688 | .188 | .022 | .0015 | 1920 | .060 |
| -7 | .4375 | .8125 | .875 | .750 | .750 | .188 | .032 | .0015 | 2110 | .090 |
| -8 | .5000 | 1.0000 | .875 | .750 | .938 | .188 | .032 | .0015 | 2350 | .120 |
| -10 | .6250 | 1.1250 | .875 | .750 | 1.062 | .250 | .032 | .0015 | 2690 | .150 |
| -12 | .7500 | 1.2500 | 1.125 | 1.000 | 1.188 | .250 | .032 | .0015 | 4480 | .210 |
| -14 | .8750 | 1.3750 | 1.125 | 1.000 | 1.312 | .375 | .032 | .0015 | 5000 | .240 |
| -16 | 1.0000 | 1.5000 | 1.125 | 1.000 | 1.438 | .375 | .032 | .0015 | 5510 | .270 |
| -20 | 1.2500 | 1.8750 | 1.375 | 1.250 | 1.812 | .375 | .032 | .0017 | 8180 | .300 |

TABLE II — Dimensions in millimeters

| Dash NO. | ø d Bore | øD Outer Ring Outside Dia. | B Over-all Width | c Outer Ring Width | ød1 Washer Outside Dia. | B1 Lubri-cation Groove Width | $r_a$1/ Fillet Max. | Radial Internal Clearance Max. | Limit Load Rating N | Mass (approx) kg |
|---|---|---|---|---|---|---|---|---|---|---|
| -3 | 4.826 | 15.875 | 15.88 | 12.70 | 14.27 | None | .6 | .038 | 3020 | .018 |
| -4 | 6.350 | 15.875 | 14.27 | 12.70 | 14.27 | None | .6 | .038 | 3910 | .011 |
| -5 | 7.938 | 17.462 | 15.88 | 12.70 | 15.88 | None | .6 | .038 | 4390 | .023 |
| -6 | 9.525 | 19.050 | 20.62 | 19.05 | 17.48 | 4.78 | .6 | .038 | 8540 | .027 |
| -7 | 11.112 | 20.638 | 22.22 | 19.05 | 19.05 | 4.78 | .8 | .038 | 9390 | .041 |
| -8 | 12.700 | 25.400 | 22.22 | 19.05 | 23.82 | 4.78 | .8 | .038 | 10500 | .055 |
| -10 | 15.875 | 28.575 | 22.22 | 19.05 | 26.97 | 6.35 | .8 | .038 | 12000 | .068 |
| -12 | 19.050 | 31.750 | 28.58 | 25.40 | 30.18 | 6.35 | .8 | .038 | 19900 | .095 |
| -14 | 22.223 | 34.925 | 28.58 | 25.40 | 33.32 | 9.52 | .8 | .038 | 22200 | .109 |
| -16 | 25.400 | 38.100 | 28.58 | 25.40 | 36.52 | 9.52 | .8 | .038 | 24500 | .123 |
| -20 | 31.750 | 47.625 | 34.92 | 31.75 | 46.02 | 9.52 | .8 | .043 | 36400 | .136 |

1/ THE CHAMFER ON BEARINGS MUST CLEAR THE MAXIMUM FILLET RADIUS GIVEN IN THE TABLE. THIS SPECIFICATION DOES NOT CONTROL BEARING CHAMFER CONTOURS.
2/ DO NOT MEASURE O.D. WHEN BEARING IS IN FREE STATE. DIMENSIONS ABOVE ARE FOR BASIC OUTSIDE DIAMETER

INACTIVE FOR NEW DESIGN

(H) DENOTES CHANGES

INCH-POUND

PREPARING ACTIVITY: NAVY-AS
CUSTODIANS: ARMY- AT NAVY- AS
AIR FORCE- 99 DLA-
REVIEW: AF-84 DLA-IS
USER:
PROJECT NUMBER: 3110-0855

MILITARY SPECIFICATION SHEET

TITLE
BEARING, ROLLER, NEEDLE-SINGLE ROW, THIN SHELL, TYPE II, ANTIFRICTION

SPECIFICATION SHEET NUMBER
MS24462 — 05 DEC 94 REV H

SUPERSEDING
MS24462 G — 17 JUN 77

AMSC- N/A — FSC 3110

DISTRIBUTION STATEMENT
A. Approved for public release; distribution is unlimited



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OMB No 0704-0188

TABLE III – TOLERANCE LIMITS — DIMENSIONS IN INCHES

| ød BASIC BORE MEAN | | ALLOWABLE DEVIATION FROM d OF SINGLE MEAN $ød_{mp}$ | | ALLOWABLE DEVIATION FROM OVER-ALL WIDTH, B | | ALLOWABLE DEVIATION FROM WASHER OUTSIDE DIA, $ød_1$ | | ALLOWABLE DEVIATION FROM LUBRICATION GROOVE WIDTH, $B_1$ | |
|---|---|---|---|---|---|---|---|---|---|
| OVER | INCL | HIGH | LOW | HIGH | LOW | HIGH | LOW | HIGH | LOW |
| .1250 | 1.2500 | +.0000 | -.005<br>-.005 | +.0000 | -.005 | +.010 | -.010 | +.0000 | -.062 |

| øD BASIC OUTER RING OUTSIDE DIA | | ALLOWABLE DEVIATION FROM D OF SINGLE MEAN $øD_{mp}$ | | ALLOWABLE DEVIATION FROM OUTER RING WIDTH, C | |
|---|---|---|---|---|---|
| OVER | INCL | HIGH | LOW | HIGH | LOW |
| .5625 | 1.8750 | +.0005 | -.0005 | +.0000 | -.010 |

TABLE IV — DIMENSIONS IN MILLIMETERS

| ød BASIC BORE MEAN | | ALLOWABLE DEVIATION FROM d OF SINGLE MEAN $ød_{mp}$ | | ALLOWABLE DEVIATION FROM OVER-ALL WIDTH, B | | ALLOWABLE DEVIATION FROM WASHER OUTSIDE DIA, $ød_1$ | | ALLOWABLE DEVIATION FROM LUBRICATION GROOVE WIDTH, $B_1$ | |
|---|---|---|---|---|---|---|---|---|---|
| OVER | INCL | HIGH | LOW | HIGH | LOW | HIGH | LOW | HIGH | LOW |
| 3.175 | 31.750 | +.0000 | -.013 | +.0000 | -.13 | +.25 | -.25 | +.0000 | -1.60 |

| øD BASIC OUTER RING OUTSIDE DIA | | ALLOWABLE DEVIATION FROM D OF SINGLE MEAN $øD_{mp}$ | | ALLOWABLE DEVIATION FROM OUTER RING WIDTH, C | |
|---|---|---|---|---|---|
| OVER | INCL | HIGH | LOW | HIGH | LOW |
| 14.288 | 47.625 | +.013 | -.013 | +.0000 | -.25 |

TABLE V OIL HOLE DATA

| BORE-DASH NO. | | NUMBER OF HOLES |
|---|---|---|
| OVER | INCL | INNER RING |
| 2 | 5 | None |
| 5 | 10 | 2 |
| 10 | 20 | 4 |

REQUIREMENTS:

1. MATERIAL: STEEL, MIL-S-8690, FED. STD. NO. 66, AISI/SAE STEEL No. 50100, 51100 AND 52100.

(H) 2. FINISH:
   A. PLATING: ZINC-NICKEL PLATE. AMS 2417, TYPE 2, OR CADMIUM PLATED IN ACCORDANCE WITH QQ-P-416, TYPE II, CLASS 2, TO A MINIMUM THICKNESS OF .0003 to .0006 INCHES.
   B. MACHINE FINISH: ANSI/ASME B46.1, SEE PROCUREMENT SPECIFICATION.

3. LUBRICANT: MIL-G-81322 OR MIL-G-23827. ALL BEARINGS SHALL BE PREPACKED WITH GREASE CONFORMING TO MIL-G-81322 UNLESS OTHERWISE SPECIFIED. ALL BEARINGS SHALL BE FILLED 80% MINIMUM WITH GREASE. IF MIL-G-23827 IS REQUIRED, ADD THE LETTER "L" AFTER THE MS21439 DASH NUMBER. IF A RELUBRICATION GROOVE IS DESIRED, ADD THE CODE "G" AFTER THE MS21439 DASH NUMBER AND BEFORE THE LUBRICANT DESIGNATOR IF APPLICABLE. MIL-G-23827 SHALL NOT BE USED FOR OPERATIONS WHERE TEMPERATURES EXCEED 250° F.

4. MARKING: THE MARKING SHALL CONSIST OF THE MS PART NUMBER AND THE MANUFACTURER'S IDENTIFICATION IN ACCORDANCE WITH MIL-STD-130.

5. PART NUMBER: THE MS PART NUMBER SHALL CONSIST OF THE MS NUMBER PLUS THE DASH NUMBER.

EXAMPLE: MS24462-106 L G

- G — RELUBRICATION GROOVE
- L — LUBRICANT DESIGNATOR
- 106 — DASH NUMBER
- MS24462 — MILITARY SPECIFICATION SHEET NUMBER

PREPARING ACTIVITY: NAVY-AS
CUSTODIANS: ARMY- AT NAVY- AS
AIR FORCE- 99 DLA-
REVIEW: AF-84 DLA-IS
USER:
PROJECT NUMBER: 3110-0855

MILITARY SPECIFICATION SHEET

TITLE
BEARING, ROLLER, NEEDLE-SINGLE ROW, THIN SHELL, TYPE II, ANTIFRICTION

SPECIFICATION SHEET NUMBER
MS24462 — 05 DEC 94 REV H

SUPERSEDING
MS24462 G — 17 JUN 77

AMSC- N/A — FSC 3110

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Form Approved
OMB No. 0704-0188

6. PACKAGING: BEARINGS SHALL BE INDIVIDUALLY PACKAGED TO THE REQUIREMENTS OF MIL-P-197. PACKAGE MARKED WITH MANUFACTURER'S NAME OR TRADEMARK, AND DATE OF LUBRICATION BY MONTH AND YEAR.

7. DIMENSIONS: ALL DIMENSIONS ARE AFTER PLATING AND ARE IN INCHES.

8. LOAD RATING: THE LIMIT LOAD RATING LISTED CAN BE DEFINED AS THE MAXIMUM RADIAL LOAD WHICH CAN BE APPLIED TO A BEARING WITHOUT IMPAIRING THE SUBSEQUENT FUNCTIONING OF THE BEARING. THE ULTIMATE OR STATIC FRACTURE LOAD RATING IS NOT LESS THAN 1.5 TIMES THE LIMIT LOAD RATING. THE LOAD RATING AS A TRACK ROLLER IS THE LOAD THE BEARING WILL CARRY AS A TRACK ROLLER FOR AN L-10 LIFE OF 20,000 REVOLUTIONS.

9. RADIAL INTERNAL CLEARANCE SHALL BE .0018 MAX.

10. TEMPERATURE: BEARING OPERATING TEMPERATURE RANGE -65° F TO 350° F.

NOTES:

1. REFERENCED DOCUMENTS ARE THE ISSUES IN EFFECT AT THE DATE OF INVITATION FOR BID.
2. FOR DESIGN FEATURE PURPOSES, THIS TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS HEREIN.

| | | |
|---|---|---|
| PREPARING ACTIVITY: NAVY-AS<br>CUSTODIANS: ARMY- AT NAVY- AS<br>AIR FORCE- 99 DLA-<br>REVIEW: AF-84 DLA-IS<br>USER:<br>PROJECT NUMBER: 3110-0855 | MILITARY SPECIFICATION SHEET<br>TITLE<br>BEARING, ROLLER, NEEDLE-SINGLE ROW, THIN SHELL, TYPE II, ANTIFRICTION | SPECIFICATION SHEET NUMBER<br>MS24462 05 DEC 94 REV H<br>SUPERSEDING<br>MS24462 G 17 JUN 77<br>AMSC- N/A FSC 3110 |
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